

Notes on Readings about Commutative Algebra

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Chapter 1

Local Cohomology and Duality

In this chapter, assume all rings are Noetherian.

1.1 Local cohomology

Let R be a commutative ring, $I \subset R$ is an ideal and M is an R -module.

Definition 1.1.1 (Local Cohomology). We define a submodule of M as

$$\Gamma_I(M) = \{m \in M \mid I^n m = 0 \text{ for some } n \in \mathbb{N}\}$$

When $\Gamma_I(M) = M$, we say M is said to be I -torsion. For any homomorphism of R -modules $\varphi : M \rightarrow N$, we have an induced homomorphism

$$\Gamma_I(\varphi) : \Gamma_I(M) \rightarrow \Gamma_I(N)$$

So $\Gamma_I(-)$ is a functor on the category of R -modules, and it can be extended to complexes of R -modules.

It is not hard to prove that $\Gamma_I(-)$ is left exact, we define

$$H_I^k(-) := R^k \Gamma_I(-)$$

the k th right derived functor of $\Gamma_I(-)$, which is called k th local cohomology with support in I .

Here are a few basic properties of local cohomology:

Proposition 1.1.2. *Let R be a ring, $I \subset R$ an ideal and M an R -module.*

(1) *One has $H_I^0(M) \cong \Gamma_I(M)$, and $H_I^k(M)$ is I -torsion for all k .*

(2) If $\sqrt{I} = \sqrt{J}$, then $H_I^k(M) \cong H_J^k(M)$ for all k .

(3) Let $\{M_\lambda\}$ be a family of R -modules. For each integer k , we have

$$H_I^k(\bigoplus_{\lambda} M_{\lambda}) \cong \bigoplus_{\lambda} H_I^k(M_{\lambda}).$$

(4) Given a short exact sequence of R -modules

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0,$$

we have a long exact sequence in local cohomology

$$\cdots \rightarrow H_I^k(M') \rightarrow H_I^k(M) \rightarrow H_I^k(M'') \rightarrow H_I^{k+1}(M') \rightarrow \cdots$$

Proof. (1) By taking an injective resolution.

(2) Note that $I \subset \sqrt{I} = \sqrt{J}$ and I is finitely generated (because R is Noetherian), we have $I^n \subset J$ for some n .

(3) $\Gamma_I(-)$ and taking homology commute with direct sum.

(4) Standard homological algebra. □

Example 1.1.3. Let p be a prime number and $I = (p) \subset \mathbb{Z}$. We compute $H_I^j(\mathbb{Z})$. Consider the injective resolution

$$0 \rightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0.$$

All $H_I^j(\mathbb{Z})$ vanishes except for $j = 1$. Note that $\Gamma_I(\mathbb{Q}) = 0$ and $\Gamma_I(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z}[p^{-1}]/\mathbb{Z}$. So $H_I^1(\mathbb{Z}) = \mathbb{Z}[p^{-1}]/\mathbb{Z}$.

1.2 Other ways to compute local cohomology

There are other ways to compute local cohomology.

Theorem 1.2.1. For each R -module M , there are natural isomorphisms

$$\varinjlim_n \operatorname{Ext}_R^j(R/I^n, M) \cong H_I^j(M)$$

for each $j \geq 0$.

Proof. For each R -module E and each positive integer n , we have a functorial isomorphism

$$\begin{aligned} \mathrm{Hom}_R(R/I^n, E) &\xrightarrow{\cong} \{x \in E \mid I^n \cdot x = 0\} \\ f &\mapsto f(1). \end{aligned}$$

□

Remark. Let $\{I_n\}_{n \geq 0}$ be a decreasing chain of ideals cofinal with the chain of ideals $\{I^n\}_{n \geq 0}$. That is, for any $n \geq 0$, there exists positive integers $p, q > 0$ such that $I^{n+p} \subseteq I_n$ and $I_{n+q} \subseteq I^n$. Recall that both can be used to compute the directed limit, so we get a functorial isomorphism

$$\varinjlim_n \mathrm{Ext}_R^j(R/I_n, M) \cong H_I^j(M).$$

We can also use the Koszul complex to compute the local cohomology. Recall that for an element $x \in R$ and $n \in \mathbb{N}$, the Koszul complex $K^\bullet(x^n; R)$ is

$$0 \rightarrow R \xrightarrow{x^n} R \rightarrow 0$$

where the nonzero term is in degree -1 and 0. We have a map of Koszul complexes $K^\bullet(x^{n+1}; R) \rightarrow K^\bullet(x^n; R)$ given by

$$\begin{array}{ccccccc} 0 & \longrightarrow & R & \xrightarrow{x^{n+1}} & R & \longrightarrow & 0 \\ & & \downarrow x & & \downarrow id & & \\ 0 & \longrightarrow & R & \xrightarrow{x^n} & R & \longrightarrow & 0 \end{array}$$

It is not hard to see that $\{K^\bullet(x^n; R)\}_{n \geq 1}$ form an inverse system. More generally, set $\underline{x}^n$ is a sequence $x_1^n, \dots, x_d^n$. The Koszul complex $K^\bullet(\underline{x}^n; R)$ is given by

$$K^\bullet(\underline{x}^n; R) = K^\bullet(x_1^n; R) \otimes_R \cdots \otimes_R K^\bullet(x_d^n; R).$$

We obtain an inverse system of complexes of R -modules

$$\cdots \rightarrow K^\bullet(\underline{x}^{n+1}; R) \rightarrow K^\bullet(\underline{x}^n; R) \rightarrow \cdots$$

Note that for every n , we have a morphism

$$K^\bullet(\underline{x}^n; R) \rightarrow R/(x_1^n, \dots, x_d^n).$$

1.3 Depth and cohomological dimension